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Factors affecting the success of accounting information system from the lens of DeLone and McLean IS model

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ABSTRACT

The main objective of this research is to investigate the impact of three different information system (IS) quality constructs, namely system quality, service quality, and information quality, as well as internal control quality, on both IS use and user satisfaction. The ultimate goal is to enhance the success of the information system. To achieve this goal, a quantitative approach using a survey questionnaire was employed to collect data from 111 respondents who are familiar with AIS use. The study utilized the Smart PLS-SEM to analyze the collected data and test the hypotheses. Additionally, the study proposed a theoretical model based on Delone and McLean's Information System (D&M IS) Success Model to support the findings. The results of the PLS-SEM analysis revealed that the paths leading from system quality to AIS utilization and information quality to AIS user satisfaction were not significant through direct correlations. However, all other links involving the IS success model were found to be significant according to the study outcomes. This research fills the gap in the limited theoretical understanding of AIS use and impact in Jordan and provides valuable contributions to existing IS literature, particularly in terms of identifying the determinants shaping the net benefits derived from AIS. The proposed model was validated in Jordanian companies implementing AIS, and the study can serve as a guide to underscore the importance of AIS. Finally, the study highlights some implications, acknowledges limitations, and presents prospects for subsequent research endeavors.

Introduction

Current organisations worldwide depend on accounting information to facilitate internal (managers) and external (investors, government bodies, financial institutions, and relevant individuals) parties for objective attainment and appropriate decision-making in economic disciplines (Kumar, Kar & Ilavarasan, 2021; Lutfi, 2021; Alshirah et al., 2021a). Although information disclosure standards frequently converge on crucial criteria for transparency, publication period, and business, investor, and other stakeholder advantages in developed nations, such circumstances remain in their preliminary stages within emerging countries resembling Jordan (Kushwaha, Kar & Dwivedi, 2021; Lutfi et al., 2022a; Tiwari & Raman, 2022). The State Securities Commission statistics reported the number of reporting and disclosure violations to be consistently above 50 % of the total number of sanctioned counterparts. Low-quality financial statements, lack of timeliness, low proactivity in information disclosure on operational situations, capital usage and corporate governance-oriented situations, erroneous financial statements, and poor corporate governance in several establishments were identified as opposed to the common ground of regional nations (Alshirah, Alshira'h & Lutfi, 2021b; Grover, Kar & Dwivedi, 2022). Specifically, big-scale businesses are limited to regulation adherence without striving to enhance the accounting information quality for high business operational efficiency (Alshirah et al., 2021b).

Accounting information facilitates investment-related decisions, efficiency assessments, and pertinent observations with alternatives to the aforementioned regulations (Lutfi, 2020; Saad et al., 2022). In line with Rommey, Steinbart, Mula, McNamara, and Tonkin (2012), information systems (ISs) should encompass gathered data, records, and processes to offer information for decision-making purposes. The multiple ISs (particularly computerised counterparts) utilised in every establishment facilitates organisations and their objective attainment. Such businesses strive towards the incorporation of IS or technologies for high optimisation within the institutional management process (Kar & Dwivedi, 2020; Lutfi et al., 2022b; Syahidi & Asyikin, 2018). In resolving most entrepreneurial concerns, ISs (Bokhari, 2005) has become one of the

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most sophisticated technologies that are necessary for businesses to promote and help mitigate various complexities (Al-Frijat, 2014). For example, Saarinen (1996) asserted IS employment to be a necessary prerequisite for success. In this vein, IS success denotes "the extent to which an information system actually contributes to achieving organisational goals, such as its effect on organisational performance" (Thon , Yap, & Raman, 1996, p. 252).

Notably, AIS denotes an IS that facilitates planning and control management through relevant and reliable information for decision-making (Alrfai et al., 2023; Lutfi et al., 2020). Based on Weygandt, Kieso, Kieso, and Kimmel (2012), the nature of AIS should constitute raw data gathering and presentation under insightful accounting information for relevant users. Specifically, AIS is pivotal for organisations to manage short-term intricacies as information is offered to complement control and monitoring functions regarding cash flow, expenditure, and costing (Idris & Mohamad, 2017) and facilitate strategic long-term planning for highly competitive and dynamic organisational settings (Idris & Mohamad, 2016). Hurt (2013) implied AIS to be a set of interdependent activities, documents, and technologies developed for information gathering, processing, and reporting to a broad group of internal and external company decision-makers. Primary AIS functions involve event or transaction documentation and information generation for performance assessment (Alshirah, Magablih & Alsqour, 2021; Alshira'h, Alsqour, Lutfi, Alsyouf, & Alshirah, 2020). The AIS enables a holistic collection and reporting of the financial transaction process (Asatiani, Apte, Penttinen, Rönkkö, & Saarinen, 2019) together with computerised-IT resource integrations to trace accounting activities.

Jordanian companies have strived towards AIS adoption for optimal organisational operations. The Jordanian government has also allocated multiple grants and initiatives to catalyse organisations that experience resource depletion following the significant rise of AIS users (Idris & Mohamad, 2016). Notwithstanding, most Jordanian organisations and other emerging nations fail to fully utilise AISs, specifically by extending the system applicability to business analytics and decision support (Lutfi, 2020), following decisional module implementation intricacies and expenditure. As such, businesses typically face challenges that influence their sustainability rate and performance. Given the organisational failure rates, which exceeds 50 % within the first few years based on the statistics in several nations (including Jordan) (Idris & Mohamad, 2017), it is deemed crucial to justify the reasons underlying organisational failure or success. A sound understanding of the reasons underpinning business success or failure is vital for organisational strength and economic equilibrium.

One of the key determinants of organisational success or failure involves inadequate financial information, accounting records, and management following past research. In other words, organisational failure is instigated by poor financial management and accounting records (Lutfi, 2020, 2022c). Based on Fitrios (2016), businesses could prevent such failure with optimal (i) documentation and financial management regardless of the underpinning causes (financing or competition) and (ii) organisational accounting system. Meanwhile, Okon, Otuza and Dada (2021) indicated that the distinction between failed and successful businesses relies on organisational approaches and effective accounting information usage.

As one of the efficient means of achieving competitive advantage and company sustainability, AIS significantly enhances and complements multiple business operation components (Aini, Anoesyirwan, & Ana, 2020; Khassawneh, 2014). Optimal AIS employment ascertains prompt financial information reporting and accurate accounting documentation for reliable financial information, informed management-level decision-making and organisational performance (Idris & Mohamad, 2017). On another note, AIS mitigates operational costs and eventually enhances profitability towards high business competitiveness and productivity (Alqudah et al., 2023; Idris & Mohamad, 2017). Overall, optimal AIS deployment could induce high system usage satisfaction for overall business operationalisation.

The primary objective of this work was to ascertain and assess the impact of AIS usage on its overall success. Given the controversies on assessment techniques and despite extensive AIS studies (Idris & Mohamad, 2017 Aini et al., 2020), this investigation encompassed multiple studies that partially illuminated the explicit relationship between AISs and relevant IS areas and AIS success (Fitrio, 2016; Ghobakhlo, Azar & Tang, 2019; Eide & Abba, 2017; Nguye & Nguye, 2020; Okon et al., 2021). The current examination aligned to the DeLone and McLean information system (D&M IS) success model (2003) to assess AIS success by measuring the AIS quality effect on AIS utilisation and user satisfaction or experience following past literature (Fadelelmoula, 2018). Specifically, the relationships between the study model latent variables were duly analysed to consider all the factors impacting decision-making quality. Although past empirical works have incorporated the D&M model for IS success measurement (Angelin, Hermawan, & Suroso, 2019; Al Fariz, Widiarto, Putri, Kusuma, & Sari, 2020; Hambal, 2020; Masung et al., 2020; Widyari, Ariyanto, Suprasto, & Suputra, 2021), scholars primarily emphasised the net benefits variable with performance as the dependent counterpart.

The present research aimed to investigate AIS implementation with the D&M model by substituting the net benefits variable to an AIS-based counterpart, such as AIS success. Additionally, the study model denoted internal control quality as an additional variable. Nevertheless, trust is a primary issue given the inadequate internal control outcomes in low profit level and cash flow from operating activities. Despite assumptions that AIS utilisation outcomes imply successful AIS, the key determinants of AIS success and how organisations could optimise AIS success remain ambiguous. The advent of AISs requires a holistic comprehension of AIS employment and the factors affecting AIS success for organisational advantages. Hence, this study seeks to fill these literature gaps by trying to respond to these questions:

RQ1: Is there any significant influence for D&M IS quality construct (system quality, service quality and information quality) on AIS usage and users' satisfaction in the listed Jordanian companies?

RQ2: Is there any significant influence for internal control quality on AIS usage and users' satisfaction in the listed Jordanian companies?

RQ3: Does AIS usage influencing users' satisfaction?

RQ4: Does AIS usage and user satisfaction affecting the AIS success in the listed Jordanian companies?

The present research expands the body of knowledge on AIS by investigating the AIS utilisation and user satisfaction antecedents along with their implications on AIS success among listed Jordanian companies. The practical contributions serve to elevate empirical knowledge on the AIS usage success assessment given the lack of studies on AIS success among listed Jordanian companies from organisational viewpoints. The remaining sections are listed as follows: Section 2 briefly elaborates on relevant literature; Section 3 presents the theoretical study framework and hypothesis formulation; Section 4 justifies the research method; Section 5 highlights data evaluation outcomes and interpretations. Notably, the final section concludes the empirical work with research discussions, limitations, and recommendations for future studies.

Literature review

The AIS success implies one of the most common dependent latent variables in AIS studies (Emad, Al-Timimi & Radhi, 2014; Ismail, 2009; Idris & Mohamad, 2017). Although Thong and Yap (1996) denotes this success as the degree to which AIS facilitates company goal attainment, debates on the study construct with the highest impact on AIS efficiency remain unresolved given the relative variance of AIS effectiveness connotation and conceptualisation in past literature (Idris & Mohamad, 2016). In this vein, AIS success has become pivotal for practitioners and academic scholars on a global scale (Marei, 2022; Teru, Idoku & Ndeyati, 2017). Despite the presence of specific elements that

potentially impact AIS success following past studies (Almaliki, Rapani, Khalid & Sahaib, 2019; Fatima, Ishtiaq, & Javed, 2021; Ganyam & Ivungu, 2019; Siqani & Vokshi, 2019), this research identified a knowledge gap in line with the literature review. It is deemed essential to understand the AIS usage impact on AIS success, particularly among listed Jordanian companies, given the scarcity of relevant research. This research intended to assess the AIS utilisation (post-adoption stages) and subsequent effects from business viewpoints.

DeLone and McLean (1992) evaluated six determinants that influenced IS success, such as "system quality, information quality, use, user satisfaction, individual impact, and organisation impact". An adjustment model was structured for IS success following Delone and McLean (2003). Although the model has become an extensively-acknowledged theoretical model in conceptual and empirical AIS success research among IS scholars (Delone & McLean, 2016; Jeyaraj, 2020; Mehta, Chauhan, & Kaur, 2021), empirical proof on the AIS success determinants remains scarce. D&M's model (12003, 992) asserted the necessity for future works to investigate other disregarded elements following the lack of (i) suitable theories to complement the research model or (ii) examination in the study area (Petter, DeLone, & McLean, 2013). Particularly, D&M (2016) emphasised the need for future research to investigate a specific IT or IS context (AIS) rather than examining general technologies or innovations. Summarily, this study has addressed the gap through D&M model expansion in an AI usage setting with an independent variable.

Theoretical model and hypothesis formulation

This section outlines the study model development, which builds on and extends the D&M success model (DeLon & McLen, 1992, 2003) following multiple assessments in various IS settings. Although most of the original D&M success model constructs were retained, a restricted set of independent variables indicates one of the success model limitations following past literature (Almaiah et al., 2022a). Future success elements, such as internal control quality (originally recommended and acknowledged as an additional quality aspect in IS success identification) were duly regarded in the research (DeLone & McLean, 2016). Such antecedents reflect several organisational elements that substantially affect AIS success.

Regarding net benefits, scholars continue regarding this variable (specifically performance) as an exogenous one despite the incorporation of D&M for IS success assessment (Angelin et al., 2019; Hambal, 2020; Al Fariz et al., 2020; Masung et al., 2020). This research aimed to investigate AIS utilisation with the D&M model by replacing net benefits with an IS-related variable: AIS success. No unified perspective or empirical study on how businesses could weigh potential AIS effects and values was determined despite much AIS research from organisational perspectives (Lutfi et al., 2022e). It is deemed pivotal to justify and authenticate business-related usefulness, feasibility, and impact although AIS presents a broad range of organisational possibilities.

Further examination should be performed to acknowledge AIS intricacies and potentialities in its success. Studies on the utilisation effect and value stage, specifically within the developing economy context remains lacking despite attempts to investigate the key determinants of AIS usage and its formal usage stage (Idris & Mohamad, 2016). A full-scale deployment at the usage stage and its subsequent implications on AIS success, particularly in listed Jordanian companies, stand out as a substantial study topic to fully comprehend the AIS usage effect. DeLone and McLean (2016) proposed the incorporation of additional aspects that were not omitted from the IS success model. As a multidimensional concept, IS success evaluation should encompass appropriate measures that complement the research objective and method, organisational environment, and assessment level parallel to DeLone and McLean (1992). The D&M IS success model was extended with constructs pertinent to the AIS domain.

Based on Fig. 1, system, information, service, and internal control quality denote individual user experience (AIS usage and AIS user satisfaction) factors where user satisfaction highlights explicit and implicit impacts on AIS success. The following section presents all the AIS success model constructs with rationales for research assumptions on the correlations between them. The study hypotheses were developed to identify the vital success elements of AIS net benefits. System usage and user satisfaction are inextricably linked in line with D&M (2003) as positive system usage experiences could impact user satisfaction and affect AIS net benefits (AIS success). Predictably, positive or negative AIS net benefits potentially influence system utilisation and user satisfaction from stakeholders' viewpoints.

The AIS quality and system use or user's satisfaction

System quality (SYQ) denotes the extent to which system functionalities could conveniently and effectively address customer requirements (DeLone & McLean, 2003). This construct constitutes the IS characteristics that generate the information: an essential success element in various settings, including AIS-based areas (Lutfi et al., 2023c; Petter, DeLone, & McLean, 2013; Tajuddin, 2015). Based on this study, system quality reliably assessed AIS success as a system with poor technical performance or incomplete technical functionalities (user interface consistency, ease of use, optimal customer needs fulfilment, response rates, and programme management) would lower user satisfaction and daily task performance following the additional time and effort required (DeLon & McLen, 2003; Wange & Lia, 2008). High system quality is a desirable IS attribute (Almaiah, Hajjej, Shishakly, Amin, & Awad, 2022b; Petter et al., 2013). Following past literature, system quality assessment positively influenced system usage in system diversity as perceived ease of use (Jaafrah, 2017; Pramesti, 2019). Although IS usage could increase through a high-quality and workable system, Ritchi, Evayanti and Sari (2020) discovered no system

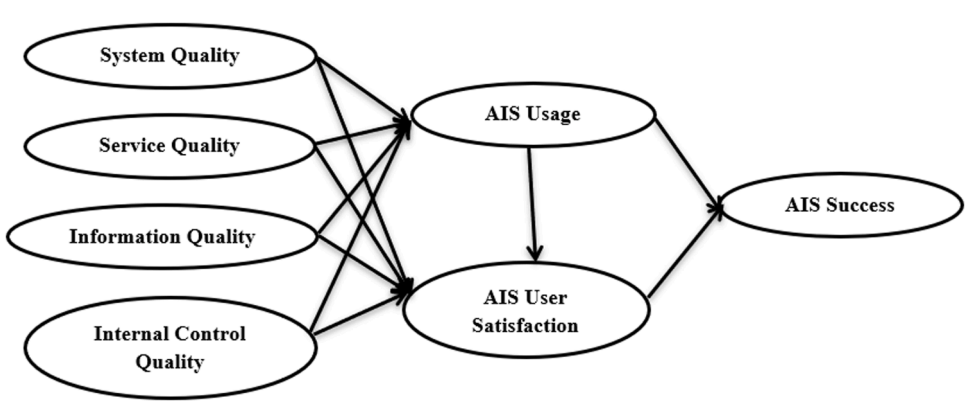

Fig. 1. Proposed model.

quality-system use correlation.

The system quality impact on satisfaction proves substantial if users experience a specific level of ease in website browsing (Ernawatiningsih & Arizona, 2021; Petter & McLean, 2009; Al-Mamary, 2019) and fulfilling anticipations on enhanced system usage attempts (Weerakkody, Irani, Lee, Hindi, & Osman, 2016). Intriguingly, the system quality impact on satisfaction would decrease if users demonstrate high self-efficacy and Internet literacy (Ernawatiningsih & Arizona, 2021). System quality is also affected by versatility, delivery, and processing and response speed in demonstrating its optimisation (Tsao, Hsieh, & Lin, 2016; Venkatesh, Thong, & Xu, 2012). Based on previous research, system quality positively influenced user satisfaction (Jaafrah, 2017; Pramest, 2019; Irawat & Syah, 2020; Wu & Wan, 2006) while other counterparts disclosed contradictory findings where system quality failed to impact user satisfaction (Wang & Liao, 2008). As such, the following hypotheses are developed:

H1: SYQ significantly influences AIS use.
H2: SYQ significantly influences user satisfaction of AIS.

The AIS service quality (SEQ) and system use or user satisfaction

SEQ denotes the extent to which a delivered service effectively addresses customer requirements in terms of system management support (DeLon & McLen, 2003). In this vein, service quality implies another IS success measurement construct (Pit, Watson & Kavan, 1995; DeLon & McLen, 2003; Lutfi et al., 2022). Arguably, companies should offer both information products and support for end-users towards IT development (DeLon & McLen, 2003; Marei, Mustafa, Othman, Daoud, & Lutfi, 2023). Although users expect an adequate level of SEQ in relevant circumstances (Almaiah et al., 2022d; Karkin & Janssen, 2014), such quality is generally assessed based on reliability, responsiveness, and assurance (Al-Khasawneh, Al-Zoubi, & Alnajjar, 2013; Barnes & Vidgen, 2006) for enhanced user relationship management and AIS success (Tajul Urus, Hasim, Syed Mustapha Nazri, & Tuan Mat, 2020; Tsao, Hsieh, & Lin, 2016). For example, high-quality service enhances satisfaction- and user-oriented perspectives and fulfils performance- and effort-based anticipations (Almaiah et al., 2021a; Marei et al., 2022; Weerakkody et al., 2016). Concerning inconsistent results on the service quality-system use correlation, some studies discovered that service quality positively impacted system usage (Jaafrah, 2017; Pramesti, 2019) while service quality proved insignificant in other counterparts (Ritchi et al., 2020; Lee & Yu, 2012).

Service quality is deemed crucial based on the following aspects that influence user satisfaction: service request response promptness, accuracy, and reliability, the preparedness to offer services, and user prioritisation (Venkatsh et al., 2012). In this vein, service quality is a user satisfaction factor given that AIS could offer enhanced and simplified services for user issue management (Tsao, Hsieh, & Lin, 2016; Venkatesh et al., 2012). Users would also reflect satisfaction if service providers offer support during system breakdown. Jaafrah (2017) contended that SEQ positively influenced users' satisfaction in the banking industry while Ernawatiningsih and Arizona (2021) claimed that SEQ affected users' satisfaction on ISs. Conversely, Rok and Al Meriou (2015) research on the industrial information system implied that service quality failed to substantially affect user satisfaction. In this regard, the following hypotheses are developed:

H3: SEQ significantly influences AIS usage.
H4: SEQ significantly influences AIS user satisfaction.

The AIS information quality (IQ) and system use or user satisfaction

IQ denotes IS output attributes (DeLone & Mclean, 2003). Specifically, accounting information quality is pivotal in organisational settings (Gackstatter, Müller-Stewens, & Möller, 2019). As information quality could optimally assess IS success in multiple settings, such as AIS-oriented domains (Urbach & Müller, 2012), this element is a valid construct in the present AIS success model. The AIS-generated information quality is highly essential at the managerial level as AIS usage relies on user information quality perspectives (Alharasis et al., 2022; Xu, 2015). The AIS capacity towards information creation, documentation, and representation implies optimised monitoring and control beyond legal organisational boundaries (Ruggeri & Rizza, 2017). In Idris & Mohamad (2017), AIS is successful with precise and prompt information that could fulfil organisational requirements to facilitate business operations and elevate user satisfaction. Idris and Mohamad (2017) and Xue, Nord, Nord, & Lin (2003) also asserted accounting information quality to be a valid source of knowledge and determined precision, promptness, completeness, and consistency as typical predictors. Based on the current study assumption, high information quality in AIS would positively induce high system utilisation.

Information quality substantially influences user satisfaction (Almaiah et al., 2022c; Raie, Lang, & Welker, 2002) based on the overall reasons underlying users' goal attainment (Chene, Jubiladoe, Capistranoe & Yene, 2015). All the computations must be accurate as system users are required to prepare error-free accounting tasks and financial statements. For example, users would be highly satisfied with using an efficient system (Chen et al., 2015) that provides complete, pertinent, dependable, responsive, and timely information (Petter & McLean, 2009). The IQ impact on user satisfaction demonstrated positive findings in some studies at IS level (Cidrali, Oliveira, Di Felice, & Aparicio, 2018; Himang, Mendoza, Manalastas, Ocampo, & Himang, 2019) while other counterparts summarised that information quality reflected inconsistent or zero effect on user satisfaction (Aparicioe, Oliveira, Bacao, & Painho, 2019; Ghobakhlo & Tangi 2015; Al-Fraihati, Joy, & Sinclair, 2019). Predictably, high information quality in AIS would positively induce high user satisfaction in line with the following hypotheses:

H5: IQ positively influences AIS usage.
H6: IQ positively influences AIS users' satisfaction.

Internal control quality and system use or user satisfaction

Internal control quality involves organisational coping mechanisms to protect and assess accounting information accuracy and reliability for operational effectiveness (Lutfi & Alqudah, 2023). Based on Tandi, Nurkholis and Subekti (2019), the internal control system reflects the internal factors impacting performance. For example, Abdelhak and Dalel (2009) revealed AIS to be an internal control and security instrument. Optimal AIS guarantees data transfer security and control within and outside the company to sustain business competitiveness by deterring data piracy and spying (Amponsah, Adekoya & Weyori, 2022). The internal control systems encompass developed procedures and policies to assure the management of its organisational goal and objective attainment (Qushtom, Al-Fasfus, Alshawahneh, & Marei, 2022). The IS application did not emphasise system usage in its implementation for e-government but focused on external and internal system success involving the society and staff, respectively. Evidently, internal control systems constitute a part of the systems package to impact users performance in e-planning applications (Alawi, Ardianto, & Sisharini, 2021).

Previous research has explored internal control in various settings. For instance, Abdallah (2013) found that IT usage had a positive influence on the internal control of the Jordanian Income Sales and Tax department, leading to increased revenue generation. Similarly, Al-Qudah's (2011) study demonstrated that AIS had a positive impact on internal control, enabling banks to generate accurate and timely information for strategic and operational decision-making. Moreover, Sajady et al. (2008) concluded that effective implementation of AIS could positively affect internal control systems, facilitating organizational financial processes and producing high-quality reports. Poor internal

control, on the other hand, could negatively impact AIS utilization. However, despite the theoretical predictions on the substantial impact of internal control quality and AIS utilization in companies presented in some empirical studies (Alodat, Al Amosh, Khatib, & Mansour, 2023; Shagari, Abdullah & Mat Saat, 2017; Shatnawi, Marei, Hanefah, Eldaia, & Alaaraj, 2022; Acklesh, Peter & Jon, 2013), there is still a lack of sufficient empirical evidence on the relationship between internal control quality and AIS usage. This study aims to contribute to the existing body of knowledge on IS and AIS by investigating the impacts of internal control quality in a specific AIS setting. In particular, the study explores how high internal control may influence users' system experience. The following hypotheses are developed to explore construct correlations:

H7. Internal control quality positively affects AIS usage.
H8. Internal control quality positively influences AIS user satisfaction.

The AIS use and users' satisfaction and AIS success (net benefits)

In IS-focused research, system use is typically defined as the level of effort expended in interacting with the system, which can be measured by the number of outputs generated by the system within a given period (Bokhari, 2005; Lutfi et al., 2023a). User evaluation plays a critical role in determining the extent to which system usage leads to improved task performance and better decision-making quality, ultimately resulting in high user satisfaction and increased system usage frequency (Bokhari, 2005; Lutfi et al., 2022b). Following Ouiddad, Okar, Chroqui, & Hassani (2020), user satisfaction could be elaborated as the degree to which users believe the system-generated information fulfil their requirements. In this vein, satisfaction is derived from user experience based on needs satisfaction, information search, and decision-making (Lutfi et al., 2020).

User satisfaction is associated with AIS usage within the AIS context (Lutfi, Al-Okaily, Alsyouf & Alrawad, 2022). Specifically, this usage supports organisational decision-making, productivity, and effectiveness (Al-Khasawneh, Malkawi, & AlGarni, 2018; Lutfi, Al-Khassawneh, Almaiah, Alsyouf, & Alrawad, 2022f; Ouiddad et al., 2020). System use encompasses user actions towards IS learning and operationalisation. System-dependent users would subsequently obtain AIS net benefits. In line with Emad et al. (2014), companies that effectively incorporate AIS tend to report positive implications on organisational performance in comparison to counterparts with restricted AIS utilisation. Similarly, Jaafrah (2017) revealed that system use positively affected the AIS net. Idris and Mohamad (2017) discovered AIS usage to have a substantial and positive impact on AIS efficiency regarding AIS success measurement. Overall, businesses should first use AIS to ascertain its effect on AIS efficiency. Past literature has highlighted the essentiality of system use as the key determinants that would allow businesses to attain favoured AIS advantages (Lutfi et al., 2022d; Tajuddin, 2015; Wang & Liao, 2008). The substantial effect of AIS utilisation on user satisfaction and AIS success proposes the following hypotheses:

H9: AIS usage has a significant influence on AIS user satisfaction.
H10: AIS usage has a significant influence on AIS success.

User satisfaction and AIS success

Although AIS success relies on users' system-oriented satisfaction (Wang & Liao, 2008), their satisfaction denotes the subjective assessment of multiple gradually-encountered and evaluated experiences (Alsyouf & Ishak, 2018; DeLone & McLean, 1992). Specifically, AIS-based satisfaction analyses users' psychological or affective states following the cognitive evaluation of their experiences with the specified system. Their AIS requirements and values need to be prioritised in system design selection and function development towards user satisfaction (Alsyouf et al., 2021; Le & Koubeki, 2010; Reddicki & Roey, 2013). For example, highly-satisfied system users reflect a positive usage experience for increased user satisfaction and net benefits. The following hypothesis is developed as user satisfaction potentially impacts net benefits (Pramest, 2019; Jaafrah, 2017; Tajuddini, 2015):

H11: User satisfaction significantly influences AIS success.

Methodology

Measurement development, study sample and data collection

For hypothesis testing, the questionnaire used in this study was originally structured in English and later translated into Arabic. The measurement items were selected based on previous research in AIS or IS, and no modifications were made to ensure their relevance in the study setting. The questionnaire was initially developed and reviewed by three experts in AIS or IS usage. A preliminary pre-test was conducted to ensure that the questions were comprehensible, clear, unambiguous, and relevant (Sekaran & Bougi, 2013). The research questionnaire was further reviewed and assessed by four senior managers and six directors employed in Jordanian organizations that utilized AIS. After the pre-test was completed, several questions were rephrased for improved readability. All items were rated on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). Table 1 presents the theoretical construct backgrounds and measurement predictors used in this research.

The study survey, which was performed for two months between 17 September and 19 November 2021, involved AIS-incorporating decision-makers. An online questionnaire was disseminated to 189 respondents who utilise AIS from the listed Jordanian companies (operating financial, service, and industrial sectors) derived from the ASE database. Notably, 111 AIS users completed the questionnaire survey. To ensure an adequate sample size, this study followed the guidelines set by Hair, Risher, Sarstedt, & Ringle (2019), which stipulate that the minimum sample size should be at least ten times the maximum number of paths leading to endogenous constructs. Therefore, a minimum sample of $N = 60$ was required for the primary outcome. However, Hair, Hult, Ringle, and Sarstedt (2021) recommended that the number of respondents should be at least eight times higher than the number of constructs in the study model, with a minimum required sample of $N = 48$. To achieve higher precision, a statistical power analysis was conducted following Cohen's (1992) approach. The analysis was performed using G*Power software (Faul, Erdfelder, Buchner, & Lang, 2009). The results indicated that a sample size of 96 respondents was necessary for an alpha level of 0.05, a moderate effect size of 0.15, and a power of 0.80. In total, 111 valid responses were obtained for SEM-PLS analysis.

Data analysis, reliability and validity results, and interpretation

To test the hypotheses, this study utilized Partial Least Squares Structural Equation modelling (PLS-SEM). This multivariate statistical technique enables the simultaneous evaluation of multiple variables within a single model (Alsyouf et al., 2022b). The method is deemed adequate for study model assessment following its capacity to evaluate

Table 1
Measurement model relevant indicators.

Construct	CA's	CR	AVEs
	> 0.700	> 0.700	> 0.500
INQ	0.820	0.874	0.582
SYQ	0.739	0.851	0.657
SEV	0.821	0.875	0.583
ICQ	0.838	0.886	0.610
AISU	0.668	0.889	0.668
AISUS	0.789	0.864	0.614
AIS SUCC	0.780	0.857	0.601

latent construct correlations through specific predictors with relatively lower sample sizes (Alshebami et al., 2023; Alshira'h, Al-Shatnawi, Al-Okaily, Lutfi, & Alshirah, 2021; Alsyouf, 2021; Hair et al., 2021; Lutfi et al., 2023b). This two-step procedure involves measurement and structural model assessment. The measurement model analyses the item and construct validity and reliability while the structural counterpart forecasts the relationship direction and significance level of the proposed correlations (Alsyouf et al., 2023, 2022b). Table 1 outlines the pertinent predictors representing the measurement model.

The collected data in this study met the reliability and validity criteria, as all scores exceeded the threshold values of 0.70, 0.70, and 0.50 for Cronbach's alpha, composite reliability, and average variance extracted (AVE), respectively, as recommended by Hair et al. (2019). The factor loadings for all items demonstrated convergent and discriminant validity, with each loading exceeding 0.40 for their respective constructs (Hair et al., 2019). Discriminant validity was further established by comparing the squared root of AVE with the construct correlation coefficients. All squared root of AVE on the diagonal line were higher than the construct correlation coefficients, indicating construct-level discriminant validity (refer to Table 2). Overall, the measurement model fulfilled the prerequisites for reliability, convergent validity, and discriminant validity at both the construct and item levels, enabling the study to proceed with structural model performance.

The structural model in this study was evaluated by applying a similar PLS-SEM algorithm. The path coefficient significance was computed with bootstrapping by generating 5000 resamples. However, Table 3 outlines the critical ratios (t-values), standardised path coefficients (β-values), and p-values (in the event of accepted hypotheses) of every recommended hypothesis. Nine out of the 11 hypotheses were accepted at 95 % and 99 % confidence level, respectively. Resultantly, system quality impacted AIS user satisfaction ($\beta = 0.142$, $t = 2.114$, $p < 0.05$), service quality affected AIS usage and user satisfaction ($\beta = 0.183$, $t = 3.217$, $p < 0.01$; $\beta = 0.175$, $t = 3.013$, $p < 0.05$, respectively), and information quality influenced AIS usage ($\beta = 0.150$, $t = 2.569$, $p < 0.05$). In this vein, H2, H3, H4, and H5 were duly supported. Regarding individual user experience, AIS usage was significantly and positively associated with AIS user satisfaction and AIS success ($\beta = 0.325$, $t = 4.825$, $p < 0.01$; $\beta = 0.227$, $t = 3.114$, $p < 0.05$). The AIS user satisfaction-AIS success connection also proved significant ($\beta = 0.380$, $t = 5.928$, $p < 0.01$) and supported H9, H10, and H11. Meanwhile, the system quality-AIS usage correlation ($\beta = 0.057$, $t = 0.944$) and information quality-AIS user satisfaction link ($\beta = 0.077$, $t = 1.121$) proved insignificant and did not support H1 and H4. The study model explanatory power is presented in Fig. 2.

Table 3
Hypotheses testing results.

Hypothesis No.	Path	β	T –Values	P - Value	Decision
H1	SYQ→AISU	0.057	0.944	0.345	NOT
H2	SYQ→ AISUS	0.142	2.114	0.035	Accepted
H3	SEQ→ AISU	0.183	3.217	0.001	Accepted
H4	SEQ→AISUS	0.175	3.013	0.002	Accepted
H5	INQ→ AISU	0.150	2.569	0.010	Accepted
H6	INQ→ AISUS	0.077	1.121	0.262	NOT
H7	ICQ→ AISU	0.325	4.825	0.001	Accepted
H8	ICQ→ AISUS	0.227	3.114	0.002	Accepted
H9	AISU→AISUS	0.137	2.020	0.044	Accepted
H10	AISU→AISSUCC	0.278	4.723	0.001	Accepted
H11	AISUS →AISSUCC	0.380	5.928	0.001	Accepted

Note: *Sig at $p < 0.05$, and ** Sig at $p < 0.01$.

Discussion

The paths from SQ to AIS usage and IQ to AIS users' satisfaction were insignificant as opposed to the remaining study correlations in the IS success model that in line with past literature (Chen, Chen, & Capistrano, 2013; DeLone & McLean, 2003, 2016; Lutfi et al., 2022c; Urbach & Müller, 2012). The AIS usage and user satisfaction relationship to AIS success were also supported (Lutfi et al., 2022a). Perceivably, the correlations depicted in the IS success model was not consistently supported (Almaiah et al., 2022b; Peter & McLen, 2009) while internal control quality for both AIS usage and AIS user satisfaction reflected significance.

Past research presented different observations on the system quality impacts on AIS usage where SQ demonstrated a more explicit effect on specified system satisfaction (Ernawatiningsih & Arizona, 2021; Saad et al., 2022; Urbach & Müller, 2012). As such, satisfaction implied an AIS usage antecedent and SQ did not directly affect AIS usage. SQ constitutes a more holistic dimension called system integration (Chen, Chen, & Capistrano, 2013; Ritchi et al., 2020). In this vein, system quality alone may be insufficient to impact AIS usage perspectives. Based on several models, information quality affected specifically information satisfaction (Alshira'h & Abdul-Jabbar, 2020; Ernawatiningsih & Arizona, 2021) and system usage (Almaiah et al., 2022a) rather than user satisfaction as a whole (system use and satisfaction together). The current study employed distinct considerations to denote satisfaction with emphasis on satisfaction based on anticipated service delivery (Alshira'h & Lutf, 2023; Alshira'h, 2019; Ouiddad et al., 2020), which potentially justifies the research outcomes. Practically, users incorporated AIS to identify accurate and useful information that fulfil their requirements and facilitate delegated task completion. The AIS usage impact was elicited with the provision of precise information, thus indicating the emphasis of information quality on AIS usage compared to AIS user satisfaction. The AIS users are also equally concerned with whether system functionalities could accelerate the overall process, hence demonstrating the focus of system quality on AIS user satisfaction as opposed to AIS usage. Resultantly, service quality denotes the only quality dimension that substantially influenced both AIS usage and AIS user satisfaction, which indicated the essentiality of service quality among other quality dimensions (Chen et al., 2013). In this regard, service quality is a vital AIS usage element.

With regards to internal control quality, the study outcomes presented sufficient proof to support the proposed hypothesis. In other words, high internal control quality reflected high AIS usage. This outcome asserts the essentiality of internal control quality as a vital management tool to protect organisational assets and regulate operational activities for high efficiency as adequate internal control is crucial to ascertain AIS usage. This outcome claims that an optimal internal

Table 2
Fornell–Larcker criterion.

	INQ	SYQ	SEQ	ICQ	AISU	AISUS	AIS SUCC
INQ	**0.835**						
SYQ	0.575	**0.763**					
SEQ	0.589	0.641	**0.775**				
ICQ	0.450	0.611	0.605	**0.781**			
AISU	0.525	0.523	0.571	0.603	**0.811**		
AISUS	0.515	0.591	0.623	0.627	0.657	**0.764**	
AIS SUCC	0.543	0.558	0.612	0.583	0.612	0.603	**0.784**

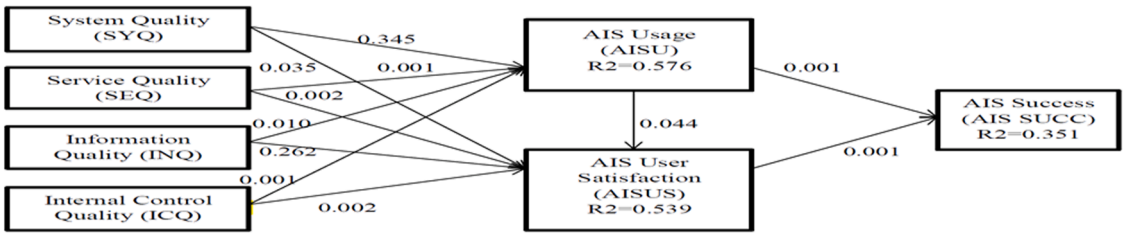

Fig. 2. Structural model results.

control systems mitigate financial irregularity and increase the possibility of AIS usage (Acklesh et al., 2013; Shagari, Abdullah & Mat Saat, 2017). Companies require an efficient internal control system for risk management and operational activity regulation following the highly competitive nature of business contexts. A high-quality internal control catalyses organisational goal attainment, protects company assets, and ascertains data relevance and integrity. The strength of internal control provides the managements with opportunities to make amendments in the wake of control weaknesses and alleviate the implications of undesirable occurrences. Despite the paucity of relevant research and association, the outcome is stimulating following its incongruence with past study assumptions (DeLone & McLean, 2016). Overall, internal control quality is an insightful and vital AIS success indicator apart from the AIS quality dimensions indicated in the model of D&M IS success.

Theoretical contributions

Theoretically, additional viewpoints were presented to assess AIS success through AIS usage and AIS user satisfaction with relevant AIS success measurement alternatives. A holistic theoretical comprehension of user behaviours within AIS is necessary by incorporating internal control quality. This study contributes to the current body of knowledge by extending the IS success model to be integrated with a vital AIS setting, which constitutes a substantial organisational portion (Daoud, 2023; Lutfi et al., 2022). In this regard, future studies could examine how internal control quality could support quality-oriented perspectives to enhance AIS usage, user satisfaction, and success viewpoints. A theoretical scenario was depicted where changes in IS users' perspectives led to intriguing discussions on which AIS quality assessments are actually valued by users in particular environments. The essentiality of AIS success was also indicated compared to net benefits as the final dependent variable. Notably, AIS success emphasizes attitudinal and behavioural development as opposed to the net benefits on functional aspects. The AIS success effectively indicates AIS performance to demonstrate user viewpoints of overall IS success in the AIS setting rather than merely highlighting its technical elements, which is generally evaluated by net benefits.

However, to achieve highly operational efficiency, organizations should not only concentrate on meeting regulatory obligations but also on constantly improving the quality and the success of their accounting information. By improving the reliability, accuracy, and timeliness of financial data, organizations can effectively optimize their operations, make more informed decisions, and identify areas for improvement. This proactive approach to accounting information success and quality is crucial for maximizing profitability, driving productivity, and maintaining businesses competitiveness in the dynamic landscape.

Practical contributions

Practically, this research indicates pivotal AIS goals. This study integrated the construct to the current model given the research scarcity involving internal control quality impacts on AIs usage and success. Internal control quality denotes a crucial AIS usage determinant in listed Jordanian companies. Notably, the study supports the incorporation of internal control quality as an AIS assessment catalyst. An optimal internal control system environment deters unfavourable practices in executing regular practices. A proper structure of internal control could safeguard AIS from complexities involving errors and fraudulence for timely and accurate information generation, strategic and operational decision-making, high productivity and financial acquisitions, and big-scale shareholder funds. Organisational managers should focus on internal control quality for a value-added business, improved company productivity and performance, and high competitiveness. Specifically, service and internal control quality should be prioritised in AIS investments by optimising information and system quality for an adequate system and internal control strategies to induce desirable user perspectives.

The results contributed to the success of IS research in the following ways: The study extended the D&M IS success model by incorporating the "internal control quality" relevant to evaluate AIS use success at the operational level. The extended model was validated in the listed Jordanian companies. Moreover, the study investigated the success of AIS in the listed Jordanian companies and found a significant and positive influence for AIS use and user satisfaction on the AIS success that was overlooked in earlier IS success evaluation models.

Additionally, the study results outlined important research implications for future practice. AIS in listed Jordanian companies efficiently use technological facilities to increase their profitability. AIS use at a large scale requires the researcher to provide IS model to evaluate the post-usage success of the technologies usage and identify various contextual factors that affect the technologies success level. Most previous research have only examined the influence of technologies use on listed Jordanian companies. The present work touched on the unexplored area by providing a validated model for assessing the post-usage success of technologies use. The management of listed Jordanian companies will be able to obtain feedback from end-users with the assistance of the model that has been proposed, compare their standard performance with their actual performance, and take appropriate corrective measures as a finding. Furthermore, the results of this study provide practitioners with a potential framework to ensure the ongoing success of integrating technologies. Specifically, the study suggests that listed Jordanian companies' management should prioritize enhancing SYQ, SEQ, IQ and ICQ to positively impact AIS success through AIS use and users' satisfaction, subsequently enhancing operational and organizational performance.

Research limitations and future research

Further empirical examination and theoretical framework assessment are necessary given that some of the hypothesised correlations were not supported by the study data. For example, system quality-AIS usage and IQ-AIS users satisfaction correlations could be further explored as the D&M model is not consistently successful in users' behaviour prediction or relationship development. Typical data collection limitations involving survey research are duly highlighted. Potential scholars could assess the study robustness across multiple settings associated with AIS usage, user satisfaction, and success. Limitations encompassing translation concerns might also arise even with

demanding translating and back-translating processes as the questionnaire was converted and translated to Arabic. It would also be interesting to widen the sample size given the considerably small number of respondents in this study. As this study was performed in Jordan, which inevitably restricts outcome generalization on a global scale, future research could resolve this limitation through large-scale samples from other emerging economies or developed counterparts. Several constructs (usage) require a specified duration for optimal measurement, which is not viable with cross-sectional research in considering responses at different intervals. This limitation is deemed common among most D&M-oriented studies. Alternatively, further research could conduct comparative research or judge AIS pre-implementation and post-usage behaviours. The system net benefits could also be examined through novel elements, such as decision-making quality or organisational performance and the integration of contingent effects as the current study only incorporated one of the novel impact factors (AIS success (effectiveness)) into the model as the net benefit of AIS usage. Finally, particular aspects (AIS training, user experience, and maturity) should be regarded to comprehend the strategic AIS advantages for decision-making support, as the selected variables were not exhaustively explored. All the aforementioned factors could determine more variables to justify actual decision-makers' behaviours. Longitudinal AIS studies should also be performed following the inconclusive outcomes in the current research.

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Declaration of Competing Interest

Author declares no conflict of interest.

Data availability

Datasets are available from the sole author upon reasonable request.

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